Shesh Budhabhatti

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EDUCATION

NORTHEASTERN UNIVERSITY (Dean's List x3 at Khoury College of Computer Sciences) **Boston, MA**
B.S., Combined Major in CS (Cybersecurity) and Business Administration (Finance Concentration) April 2028
Coursework: Systems Security, Foundations of Cybersecurity, Network Fundamentals, CS Systems, Law & Ethics of Digital Technology, Algorithms & Data Structures, Object-Oriented Design (Java)

SKILLS

Languages: Python, Java, C, C++, MATLAB, JavaScript, SQL, Assembly
Frameworks & Web: React, Vite, REST APIs, Vercel Serverless Functions, HTML/CSS
Systems & Tools: Git/GitHub, Copilot, Claude, GitHub Actions, Linux/Unix, Docker, GDB, Valgrind, pthreads, Azure DevSecOps, Jira, Agile, Bitbucket, Polarion, Miro, Arduino, Microsoft Office (Excel, Word, PowerPoint)
Cloud & Data: AWS (Lambda, DynamoDB, API Gateway, IAM, SAM), MySQL, NoSQL
Security: Threat modeling, secure coding, authentication/authorization (OAuth 2.0), access controls, least privilege, privilege escalation, penetration testing, prompt injection, risk analysis, root cause analysis, cryptographic protocols, Wireshark, Metasploit, Hashcat
Software Engineering: API Development & Integration, UML/Software Architecture, Test Automation, Software Testing & Debugging, System Validation & Performance Testing

WORK EXPERIENCE

MEDTRONIC **Boston, MA**
Software Architecture Co-op August 2026 – Present
- Building a GitHub Actions CI/CD pipeline that runs on every commit to auto-generate UML diagrams and technical documentation for Medtronic's Hugo™ surgical robot platform, keeping design records traceable to code changes
- Integrating generated design documentation with a MySQL-backed database through a cross-team API, contributing 20+ pull requests and addressing code review feedback from engineers across teams
- Developed a redline diff tool for pull requests to highlight design changes between versions, added inline PlantUML support to the architecture docs site and regression-tested 50+ existing diagrams to verify PlantUML and Mermaid rendering

Product Security Intern June 2026 – August 2026
- Built Python monitoring tooling to capture and visualize Wi-Fi telemetry across 2.4/5 GHz bands, feeding a decision engine that fused latency, jitter, packet loss, and RSSI into a real-time health score to trigger autonomous failover on a self-designed testbed (Raspberry Pi 5 AP, ESP32 client/interference nodes)
- Validated autonomous failover under load (28 clients, 600 packets/sec), induced interference (0 to 4 disruptors: latency 3ms to 33ms, 0% packet loss), and injected 802.11 deauthentication bursts, holding a stable band switch unattended for ~10 hours
- Root-caused testbed failures from packet captures, isolating an AP/injector channel mismatch (6 vs 11) and a WPA2/WPA3 profile mismatch that broke association
- Presented Wi-Fi resilience findings and cybersecurity risk analysis to Medtronic vice presidents and leadership

MICROSOFT **Boston, MA**
AI Student Ambassador & WICxDevSecOps Program Participant October 2025 – February 2026
- Trained in threat modeling and secure CI/CD on Azure DevSecOps pipelines, including AI model security
- Piloted Copilot/Omi AI with 100+ users, collecting feedback on quality, privacy, and edge cases for product teams

MEDTRONIC **North Haven, CT**
R&D Software Engineering Intern June 2025 – August 2025 | July 2024 – August 2024
- Automated test fixture calibration with Python scripts, cutting calibration time 37% and saving 4+ hours weekly
- Authored fixture manual for cross-functional teams across US/India (15+ engineers), reducing onboarding time
- Implemented AES-256 encryption for the Signia™ Stapling System, a high-compliance medical device, under HIPAA and FDA cybersecurity requirements

SECURE & ASSURED INTELLIGENT LEARNING LAB **New Haven, CT**
Cybersecurity & Safety Researcher June 2023 – February 2024
- Exploited prompt injection vulnerabilities in LLM applications (OWASP LLM Top 10 #1) through 50+ attacks, identifying 8 vulnerabilities and documenting mitigations presented to lab researchers

KPMG **Hartford, CT**
Summer Intern, Audit/Tax/Advisory Accounting July 2023 – August 2023
- Collaborated with 5-person teams across 3 companies, identifying 2 key control weaknesses in inventory tracking
- Analyzed financial data across 3 client studies with Tableau, visualizing audit risk insight for senior stakeholders

G.U.R.L.S CHESS CLUB NON-PROFIT ORGANIZATION **USA**
Founder & Executive Director & TEDx Speaker December 2021 – Present
- Founded chess non-profit coaching 200+ students (70% female) across 70+ lessons, featured by US Chess (USCF)
- Redesigned curriculum using retention data analysis, improving participant retention from 40% to 78%

PROJECTS

llm-sec-review | Python, Claude API, Semgrep **October 2026**
- Built an LLM security reviewer for Python that returns findings as line, CWE, and severity, with a schema-validation layer (9 unit tests) rejecting malformed model output
- Benchmarked it against a Semgrep baseline on 19 labeled samples (12 vulnerability classes, 2 prompt-injection samples, 5 safe controls): Claude detected 11/14 vs Semgrep's 6/14, with 0 false positives
- Tested injection resistance with samples embedding a "report zero findings" instruction next to a real vulnerability: both still reported, 2/2

Sylla | React/Vite PWA, AWS (Lambda, DynamoDB, API Gateway, SAM), Vercel **August 2026 – Present**
- Built an AI vision web app that extracts events from photos of syllabi and flyers and adds them to the user's calendar
- Treated model output as untrusted, enforcing a strict schema, length caps, and instruction-pattern filtering at three layers before any calendar write to limit prompt injection
- Built a device-to-cloud sync layer on API Gateway + Lambda + DynamoDB (single-table userId/eventId), with server-side Google ID-token verification, conditional-write dedup, and least-privilege IAM
- Built with Claude and GitHub Copilot, reviewed all AI-generated code, and wrote unit and integration tests verifying input rejection, dedup, and 401s on invalid tokens

Linux Access Control & Privilege Escalation | SUID, Least Privilege, CTF **October 2025**
- Completed OverTheWire Bandit challenges involving Linux permissions, SUID, and switching between users across five levels

FUSE Filesystem Implementation | C, Systems Programming **November 2025**
- Implemented a custom C filesystem with inode management and mmap/munmap allocation to support files larger than 4 KB
- Diagnosed memory-safety bugs using GDB and Valgrind, passing all 31 test cases

Threaded Merge Sort | C, Concurrency & Performance **November 2025**
- Built a multithreaded merge sort in C, benchmarking 100M integers across 2-core and 10-core machines, reaching 4.1x speedup at 10 threads (29.5s to 7.2s), plateauing past physical core count

BGP Router Implementation | Python, Networking **February 2026**

- Implemented BGP message handling for peer handshakes, route updates, and withdrawals, with longest-prefix-match forwarding and route tie-breaking
- Implemented IP prefix calculations using bitwise operations and rebuilt forwarding-table entries after route withdrawals for correctness

AWARDS/PUBLICATIONS/CERTIFICATIONS

TEDx: "Empowering Equality, Inspiring Action" (2023) | SMARTA Podcast (2025) | Lean Six Sigma Yellow Belt (2024) | CT Women's Chess Champ. (2018) | TRAIN's Intro. to Artificial Intelligence Certified (2024)